

ASSIGNMENT #02

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Friday, February 13th TA's Mailbox

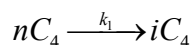
INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Plant Wide Process Control.

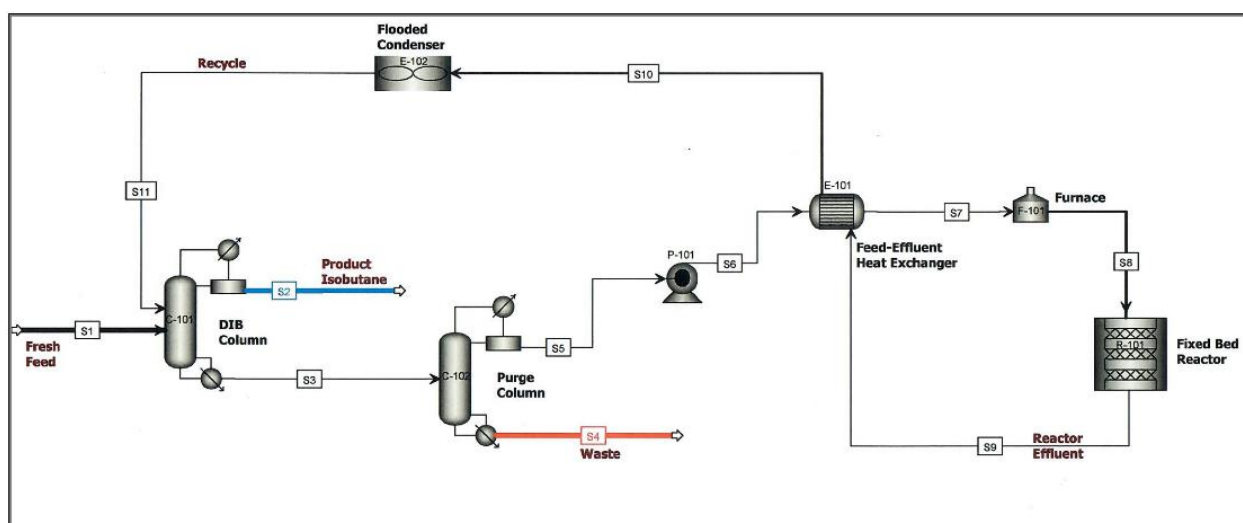
A process flowsheet diagram for the production of iso-butane via the isomerization of n-butane is shown in the Figure below: The isomerization reaction takes place in the fixed bed reactor with pressure drop equal to 15 psi. The kinetics of the reaction are shown below:



$$R_1 = k_{10} \exp\left\{-\frac{E}{RT}\right\} C_{nC_4}$$

$$E = 30,000 \text{ Btu} / \text{lbmol}$$

$$k_{10} = 4 \times 10^8 \text{ l} / \text{hr}$$



To better understand the process, open the attached ASPEN simulation and carefully study the operating conditions of all unit operations. The distillate product from the DeIsoButanizer column (DIB) has a specification of 2 mol% n-butane and is directed to a storage tank. The plant uses cooling water from near-by cooling tower designed with 75% efficiency with hot water inlet temperature of 120 °F and air ambient temperature of 90 °F with relative humidity 70%. A steam plant from a nearby facility supplies saturated steam (with quality 0.95) at three different levels of pressure: low pressure at 50 psig, medium pressure at 150 psig and high pressure at 600 psig. Natural gas from an existing pipe line is used for the fired heater that heats the feed to the reactor. The heat transfer coefficients in all the heat exchangers can be taken as the middle values of the ranges that are listed in the **CHE Calculator**.

Assume that the pressure drop in the heat exchanger is 10 psi both on the tube side and 5 psi on the shell side and the pressure drop in the fired heater is 60 psi. The pressure drop in the liquid line from the pump discharge to the inlet of the fired heater is approximately 8 psi.

This assignment has three parts:

Part A: Flowsheet Modifications

1. First you must modify the simulation to obtain the product specifications of 2 mol% n-butane in the tops of C-101, which is done through creating a design specification in the column by varying the boil-up rate, through that, the diameters of the distillation columns can be found. This will help calculate the height of the liquid in the column sump. Some modifications will also be needed to accommodate the additional information on pressure drop given in the problem description above.
2. Second you must create three utilities (cooling water, steam and natural gas) and attach them to the appropriate unit operations. Utility consumption will be used to calculate the steady-state values of the manipulated variables for the process control loops that you will identify in **Part B**.

Part B: Identification and Design of Process Control Loops

Following the procedure that I have outlined in the lectures, calculate the Control Degrees of Freedom for this process and design all process control loops using the additional information outlined below.

Data and Assumptions

1. Both refluxes in the **DIB** column and the **Purge** column are on flow control
2. Residence time in the reflux accumulation drums: 4 **min**
3. Residence time in the distillation column sumps: 2 **min**
4. The waste stream is also on flow control due to environmental restrictions
5. Neither the fresh feed nor the product flow rates are fixed by other plant considerations
6. Heating value of natural gas used in the fired heater: 50.0 **MJ/kg**
7. The heat capacity of water can be taken equal to 4.18 **kJ/(kg C)**
8. Temperature rise in the cooling water is limited to 11 °C

Process Variables Units for Reporting

1. Liquid flow rates must be given in **gpm**
2. Gas flow rates in **lb/hr**
3. Temperature in **deg F**
4. Pressure in **psia**
5. Height in **ft**

For each control loop you have identified write a summary similar to the template shown below:

Template: Design of Process Control Loops

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	DIB Column Reflux Flow	Reflux Flow		Reflux Flow	
2					

...

Part C: Process Simulation Calculations

From pilot plant studies it was recommended to increase the reactor feed temperature by 10 °F in order to increase production rate. Change the set point of the temperature controller and answer the following questions after carrying out the simulation with the new reactor feed temperature.

1. Briefly explain your logic in setting up the simulation for the change in the temperature loop set point value from 390 deg F to 400 deg F. Did you have to change the set points of any other controllers? If yes provide a brief explanation as to why and then enter their new values in the template shown below.
2. Run the simulation in order to calculate the new steady-state values of the manipulated values for all controllers and enter the new values in the template shown below

Loop #	Controller Description	Controlled Variable: CV	Old Set Point: SP	New Set Point: SP	MV	Old Steady State Value	New Steady State Value
1	DIB Column Reflux Flow	Reflux Flow Rate			Reflux Flow Rate		
2							
...							

Template: Simulation Calculations after Set Point Change

Part A

The objective of this assignment is to develop a plant-wide process control strategy for the production of iso-butane through the isomerization of n-butane. The goal is to design and implement appropriate control valves and control loops to maintain key process variables at their desired set points. Emphasis is placed on selecting suitable flow variables as manipulated variables to ensure stable and efficient operation of the process.

- 1) After Following the instructions above on applying pressure drop in the equipment, and providing a product specification for the columns the diameter of the columns can now be obtained.

The height of the liquid sump can be calculated using the equation:

$$H = \frac{\dot{V}}{A} * t_r$$

Where $\dot{V}$ is the volumetric flow rate leaving the bottoms stream of the column, A is the area of the cross section of the column, $A = \frac{\pi D^2}{4}$, and t_r is the residence time of the sumps in the columns given to be 2 minutes.

The calculations are summarized in the table below along with the sump level for both columns:

Table 1- Height Calculation of Sump Level in C-101 & C-102

Parameter	C-101	C-102
Vflow, Cuf/min	26.73	1.05
V, Cuf	53.46	2.09
D, ft	9.50	5.50
A, ft ²	70.88	23.76
h, ft	0.75	0.09

- 2) For Utilities, cooling water, all the types of steam, and natural gas were added with the specifications that were given in the problem statement.

Starting with cooling water. It was given that the dry air temperature is 90 °F with a humidity of 70%. Using the WETBULBTEMP() function in the CHE calculator and inputting the air temperature and humidity level, the inlet of the cooling water to our plant will be 81.8°F. Given that the temperature rise of cooling water is 11 degrees Celsius, the outlet temperature of the cooling water will be 102°F. With these temperatures and the providing the cooling value of - 4.18 kJ/kg, a cooling water utility was added in the simulation.

For steam, the respective pressures were added for all the types of steam along with the quality of the inlet of the steam to 0.95 and the outlet to be 0.

For Natural gas the heating value of natural gas in the fired heater was input as 50 MJ/kg.

Cooling water will be added as a utility in the condenser of both distillation columns C-101 and C-102 and added in the flooded condenser E-102. LP steam was added to the reboiler in C-101, while MP steam was added to the reboiler in C-102. Finally, natural gas will be added in the fire heater upstream to the adiabatic reactor.

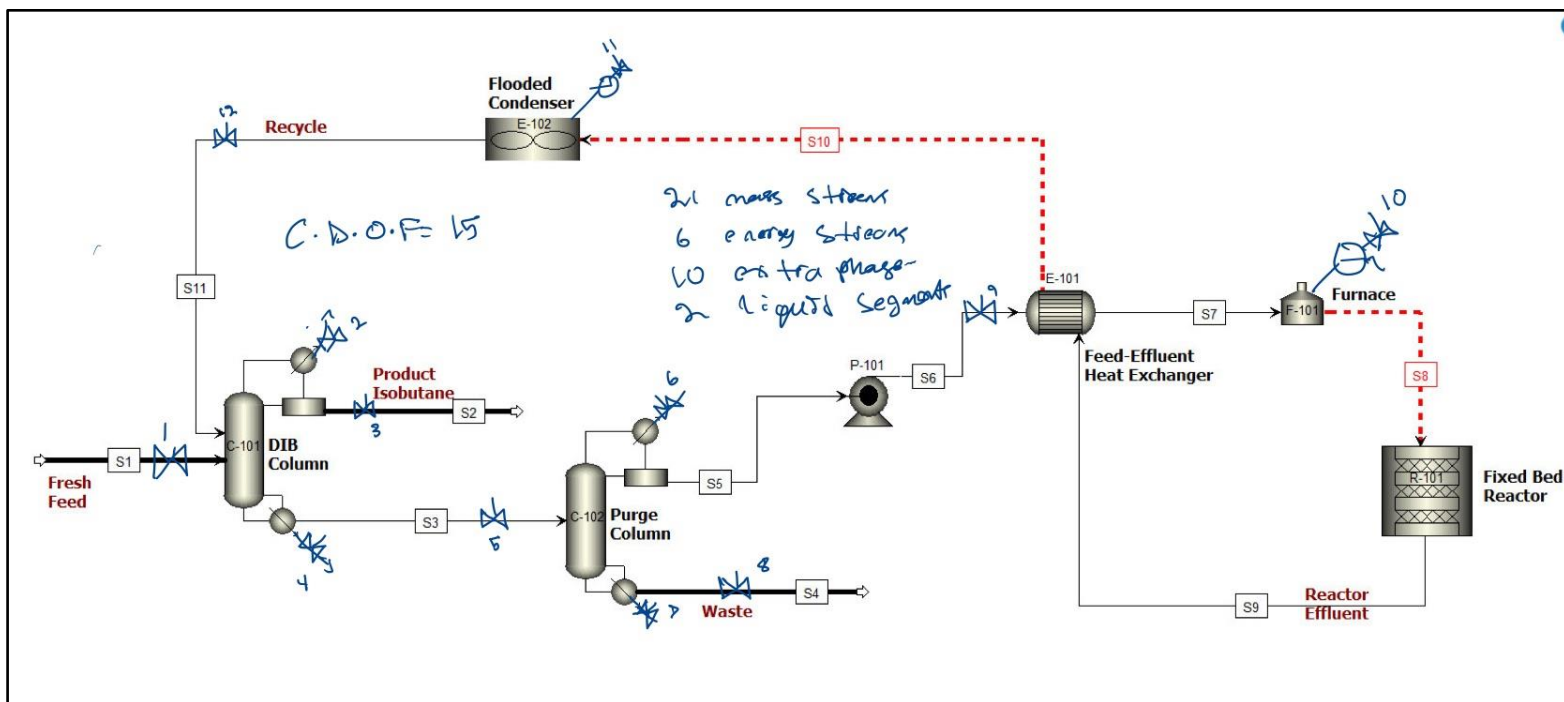


Figure 1- PFD with Valves

Part B

The plant-wide control design begins with an evaluation of the available control degrees of freedom (DOF). These were determined by identifying all independent manipulated inputs in the process and then subtracting the constraints introduced by material balances, phase behavior, and equipment characteristics. The flowsheet contains 21 material streams (including internal column streams), 6 energy interactions associated with utilities and heat duties, 10 additional phase relationships created by vapor-liquid transitions in separation equipment and heat exchangers, and 2 liquid transport segments that do not provide inventory (the pump and the liquid-side portion of the heat exchanger). Following the methodology presented in Chapter 2 of Introduction to Process Control, particularly for multivariable systems such as distillation columns, the overall number of controllable degrees of freedom was determined to be 15.

$$C.D.O.F. = 21 + 6 - 10 - 2 = 15$$

This means that, in theory, up to fifteen independent manipulated variables could be implemented without over-specifying the process. In practice, however, not all of these handles are equally suitable. Some streams are already assigned fixed roles by the problem statement (for

example, both reflux streams must operate under flow control), and certain valve placements are avoided to ensure safe and reliable plant operation.

The control structure is therefore organized around fundamental operational requirements: inventory regulation, pressure stability, temperature control, and product specification compliance. For both distillation columns, liquid inventories must be maintained in the bottom sump and in the reflux accumulator to prevent flooding or dry-out. Level control is achieved by manipulating the corresponding outlet liquid flow streams. Column pressure must also be regulated because it directly influences vapor–liquid equilibrium and separation efficiency. Since both columns use total condensers, pressure control is most effectively implemented by adjusting condenser cooling-water flow, which directly changes the condensation rate and vapor holdup at the column top.

Because reflux flows are already specified to be on flow control, they cannot be used as manipulated variables for other objectives such as composition or level regulation. This restriction requires the use of alternative handles. Steam flow to the reboiler serves as the primary manipulated variable for influencing separation strength and maintaining product composition. In cases where inventory control is required, fresh feed or product streams can be used as level-control handles because they are not externally constrained by other plant requirements.

The reactor and heating sections introduce additional regulatory needs. The reactor inlet temperature must remain tightly controlled to ensure stable fixed-bed performance and consistent conversion. This is accomplished by manipulating natural gas flow to the fired heater. Reactor feed flow is controlled using a valve located downstream of the pump, where the lower volumetric flow simplifies valve sizing and improves controllability. In the recycle and condensation section, the cooling-water flow to the flooded condenser is used to regulate recycle temperature, while a recycle-line valve provides a practical means of influencing recycle rate and overall pressure behavior within the reactor loop.

Overall, the proposed control configuration balances theoretical controllability with practical considerations. The selected manipulated variables maintain stable operation, satisfy environmental and product-quality constraints, and respect the imposed limitations on reflux flow control. The resulting control structure ensures that the process operates safely and consistently while utilizing only the available and appropriate degrees of freedom.

Below in the table, the 12 control and manipulated values are shown along with the setpoints and steady-state flows. The flows for the water in gpm are calculated using the density at 90 F to be 8.32 lb/gal.

Table 2- Design of Process Control Loops

Loop #	Controller Description	Controlled Variable: CV	Set Point Value: SP	Manipulated Variable: MV	Steady-State Normal Value:
1	C-101 Feed Flow	Sump Level	0.75 ft	Feed Flow	120.91 gpm
2	C-101 CW Flow	Column Pressure	100 psia	Cooling Water Flow	52,691 gpm
3	C-101 Reflux Flow	Accumulator Level	13.5 ft	Reflux Flow	122.69 gpm
4	C-101 Steam Flow	n-butane Purity	0.02	Medium Steam Flow	56,142 lb/hr
5	C-102 Feed Flow	Sump Level	0.09 ft	Feed Flow	199.96 gpm
6	C-102 CW Flow	Column Pressure	65 psia	CW Flow	14833 gpm
7	C-102 Steam Flow	Accumulator Level	4.001 ft	LP Steam Flow	15125 lb/hr
8	C-102 Bottoms Flow	Waste Flow	7.82 gpm	Bottoms Flow	7.82 gpm
9	E-101 Cold Inlet Flow	Cold outlet Temperature	280 F	Cold Inlet Flow	185.61 gpm
10	F-101 Fired Heater Flow	Reactor Inlet Temperature	390 F	Natural Gas Flow	321.86 lb/hr
11	CW Flow in Flooded Condenser	Temperature In E-102	125 F	CW Flow E-102	8967 gpm
12	E-102 outlet Recycle Flow	Pressure in R-101	572 psia	E-102 Recycle Flow	190.68 gpm

Part C

To evaluate the effect of increasing production rate, the setpoint of the reactor inlet temperature controller was increased from 390°F to 400°F. The fired heater natural gas flow serves as the manipulated variable for this loop. When the setpoint was raised, Aspen increased the natural gas flow from 321.86 lb/hr to 335.17 lb/hr in order to achieve the new reactor inlet temperature of 400°F. No other controller setpoints were intentionally changed; all other control loops were allowed to respond naturally to the new steady-state condition.

Increasing the reactor inlet temperature increases the reaction rate, which affects reactor conversion and downstream compositions. As a result, several manipulated variables adjusted automatically to maintain their controlled variables at their original setpoints.

In the DIB column (C-101), the n-butane purity specification remained fixed at 0.02. However, because the reactor conversion increased, the required separation duty changed. The steam flow to the reboiler decreased from 56,142 lb/hr to 44,806 lb/hr. This indicates that less reboiler duty was required to maintain the same product specification under the new feed conditions.

Cooling water flows in both columns also shifted slightly to maintain pressure control. For example, the C-102 condenser cooling water flow decreased slightly (14,833 gpm to 14,767 gpm), while the flooded condenser cooling water flow increased (8,967 gpm to 9,196 gpm) to maintain recycle temperature and pressure stability. These changes reflect the redistribution of heat duties throughout the process due to the higher reactor temperature.

Inventory control loops (sump and accumulator levels) remained stable with only minor adjustments in flow rates. The level setpoints were not changed; instead, the corresponding manipulated flows adjusted slightly to maintain material balance under the new steady state.

Reactor pressure (572 psia) remained constant, demonstrating that the pressure control loop successfully compensated for the increased vapor generation caused by the higher reactor temperature. The recycle flow increased slightly (190.68 gpm to 191.64 gpm), indicating a small shift in overall recycle dynamics.

Overall, the primary direct effect of increasing reactor feed temperature was an increase in fired heater duty. The downstream columns and condenser systems adjusted their energy and flow rates automatically to maintain pressure, level, and composition specifications. No additional controller setpoints required modification, as all control loops were capable of maintaining stability under the new operating condition.

Table 3- Simulation Calculations after Setpoint Change

Loop #	Controller Description	Controlled Variable: CV	Old Set Point: SP	New Set Point: SP	MV	Old Steady State Value	New Steady State Value
1	C-101 Feed Flow	Sump Level	0.75 ft	0.65 ft	Feed Flow	120.91 gpm	120.91 gpm
2	C-101 CW Flow	Column Pressure	100 psia	100 psia	CW Flow	52,691 gpm	52,691.4 gpm
3	C-101 Reflux Flow	Accumulator Level	13.5 ft	11.9 ft	Reflux Flow	122.69 gpm	122.69 gpm
4	C-101 Steam Flow	n-butane Purity	0.02	0.02	Steam Flow	56,142 lb/hr	44,806 lb/hr
5	C-102 Feed Flow	Sump Level	0.09 ft	0.10 ft	Feed Flow	199.96 gpm	200.17 gpm
6	C-102 CW Flow	Column Pressure	65 psia	65 psia	CW Flow	14,833 gpm	14,767 gpm
7	C-102 Steam Flow	Accumulator Level	4.001 ft	4.01 ft	Steam Flow	15,125 lb/hr	15,061 lb/hr
8	C-102 Bottoms Flow	Waste Flow	7.82 gpm	7.82 gpm	Bottoms Flow	7.82 gpm	7.82 gpm
9	E-101 Cold Inlet Flow	Cold Outlet Temperature	280 F	280 F	Cold Inlet Flow	185.61 gpm	185.83 gpm
10	F-101 Fired Heater Flow	Reactor Inlet Temperature	390 F	400 F	Natural Gas Flow	321.86 lb/hr	335.17 lb/hr
11	CW Flow Flooded Condenser	Temperature in E-102	125 F	125 F	CW Flow E-102	8967 gpm	9196 gpm
12	E-102 Outlet Recycle Flow	Pressure in R-101	572 psia	572 psia	E-102 Recycle Flow	190.68 gpm	191.64 gpm

Below is a table that summarizes the calculations for the accumulator drums height and the sump level height in the columns for old and new steady-state values:

Table 4- Accumulator Height Calculations

Accumulator	C-101 Old	C-102 Old	C-101 New	C-102 New
Disillate mole flow, lbmol/hr	549.52	866.10	549.52	866.10
density, lbmol/cuft	0.56	0.59	0.56	0.59

relux ratio	10.37	0.80	8.03	0.80
V reflux rate, cuft/hr	10,209.90	1,180.67	7,905.93	1,181.96
V reflux rate, gpm	1,272.92	147.20	985.67	147.36
Vreflux, cuft	680.66	78.71	527.06	78.80
D, ft	8.00	5.00	7.50	5.00
A, ft ²	50.27	19.63	44.18	19.63
h, ft	13.54	4.01	11.93	4.01

Table 5- Sump Level Calculations

Sump Level	C-101 Old	C-102 Old	C-101 New	C-102 New
Vflow, Cuf/min	26.73	1.05	26.76	1.05
V, Cuf	53.46	2.09	53.52	2.09
D, ft	10.25	5.25	9.13	5.24
A, ft ²	82.45	21.66	65.54	21.56
h, ft	0.65	0.10	0.82	0.10

References

- 1- Caracotsios, M. C. (2023). CHE Calculator (Version 14) [Computer software]. Department of Chemical Engineering, University of Illinois Chicago.
- 2- Aspen Technology, Inc. (2020). Aspen Plus (Version 14) [Computer software]. AspenTech. <https://www.aspentech.com>
- 3- Caracotsios, M. (Year). Introduction to chemical process control (Chapter 2) [Lecture notes]. Department of Chemical Engineering, University of Illinois Chicago.